Owen Vellegas

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Education

University of Washington **Seattle, WA**

Bachelor of Science in Computer Science *Sep 2024 - June 2028*

- Relevant Coursework: Machine Learning, Deep Learning, Natural Language Processing, Computer Vision, Data Structures and Algorithms, Discrete Mathematics, Probability and Combinatorics, Operating Systems, Distributed Systems, Artificial Intelligence

Experience

Amazon June 2026 – Aug 2026

Software Engineer Intern *Bellevue, WA*

- Built sub-minute live order tracking for Amazon Quick Commerce using SQS, EventBridge, and Step Functions
- Engineered per-order refresh schedules scoped to the delivery lifecycle, removing compute spend on inactive orders
- Reduced update latency 75% and cut billing cost $14,400/month by moving tracking to express state machines

Fred Hutch Cancer Research Center Sep 2025 – May 2026

Machine Learning Research Intern *Seattle, WA*

- Developed predictive models helping doctors prevent skeletal complications in metastatic bone cancer patients
- Engineered clinical features to uncover patterns automating risk assessment across 30,000+ patient records
- Enabled collaborative prediction across 3 cancer centers through federated learning, achieving 0.815 AUC

University of Washington NLP Lab Jan 2026 – May 2026

Undergraduate Researcher *Seattle, WA*

- Built and shipped a AI tool enabling Innocence Project attorneys to extract case timelines from court transcripts
- Designed event storage retrieval to preserve context during processing, improving date extraction accuracy by 30%
- Automated timeline evaluation against attorney-written references, benchmarking extraction quality across 50+ cases

Scale AI Feb 2025 – May 2025

Software Engineer Intern *San Francisco, CA*

- Constructed scalable RLHF pipelines on Kubernetes, accelerating preference model training and policy alignment
- Developed a data labeling feedback API to capture rater uncertainty, exposing training signals for data refinement
- Contributed to an LLM evaluation suite (HFC) to benchmark models on safety, factuality, and reasoning failures

Projects

Hierarchical Reasoning Model Evaluation (ARC-AGI)

- Built an ARC-AGI evaluation pipeline comparing HRM and Transformer models on visual puzzles
- Matched models at 27M parameters and preserved answer refinement for controlled comparisons
- Varied reasoning steps and data augmentation to isolate what improved prediction accuracy
- Found a simpler Transformer achieved accuracy within roughly 5 percentage points of HRM on ARC-AGI-1

LLM Reasoning Optimization (s1 Reproduction)

- Built an evaluation pipeline to test whether longer reasoning improves LLM math accuracy
- Implemented budget forcing to extend reasoning before final answers without retraining
- Compared accuracy across token budgets to measure trade-offs between quality and compute
- Reproduced an AIME24 accuracy increase from 50% to 57% by extending model reasoning

Skills

Languages: Python, TypeScript, Java, C++, SQL, HTML/CSS

Frameworks: PyTorch, scikit-learn, Hugging Face Transformers, NumPy, pandas

Libraries & Tools: Git, Docker, Linux, AWS, vLLM, Weights & Biases